

JEAN-MILAN ALBAREDE

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EDUCATION

Purdue University, College of Engineering, West Lafayette, Indiana

May 2028

Bachelor's degree, Electrical and Electronics Engineering

GPA: 3.96

Relevant Coursework

Differential Equations, Linear Algebra, EE Fundamentals I, Intro to Digital System Design, Physics I & II, Advanced C

SKILLS

Software: Altium Designer, LTspice, Git/GitHub, Linux, Microsoft Excel, PowerPoint, SolidWorks, Fusion 360

Programming: Python, C, Verilog, Java, MATLAB, LaTeX

Language: Fluent in Russian, French, and English

Other: Data analysis w/ Python, Soldering, Oscilloscope, Signal Generator, I2C, UART, Self-Motivated

PROFESSIONAL EXPERIENCE

FPGA & Hardware Design Intern - BaPSF, UCLA

Summer, 2026

- Designed an **FPGA**-based 15-channel nanosecond pulse generation system on a **Xilinx Artix-7**, producing user-adjustable 45–55 ns pulses with 0.5 ns resolution for stark spectroscopy experiments.
- Implemented **dynamic MMCM reconfiguration** and **clock-domain synchronization** logic enabling runtime adjustment.
- Designed a custom 4-layer high-speed PCB, in **Altium Designer**, featuring length-matched, impedance-controlled routing, 15 gate-driver channels, and SMA outputs for low-jitter pulse distribution.
- Validated timing performance through Vivado simulation and oscilloscope measurements.

Engineering Intern - N6 Tec, Biotech Startup

Summer, 2023

- Created part models in **SolidWorks**, applying **GD&T** tolerances; received feedback through iterative design reviews and markups.
- Prototyped and fabricated small components that supported company projects utilizing additive manufacturing.

ENGINEERING ORGANIZATIONS

FPGA Basketball Game Controller, SoCET Intro I

Spring, 2026

- Led top-level FPGA integration, combining RTL modules developed by team members into a cohesive, fully functional system while resolving interface, timing, and system-level issues.
- Designed **SystemVerilog** modules and **testbenches** for button conditioning, timing generation, display control, and system-level operation.
- Collaborated on FPGA validation in **Xilinx Vivado**, integrating logic with a **custom PCB** for display drivers & user input.

Embedded Systems @ Purdue, Harmonicore Project (FPGA Subteam)

Spring, 2026

- Converted Python audio-processing algorithms into **RTL** diagrams and **Verilog** for an **Artix-7 FPGA**, implementing live effects like echo, chorus, or reverb on analog audio input (via **Xilinx Vivado**).

IEEE MTT-S, Microwave Theory & Techniques Society (Member)

Fall, 2025

- Aided in designing the transceiver's **signal chain** for both transmit (Tx) and receive (Rx) paths via the evaluation of RF front-end components (PLLs, LNAs, Mixers, Phase Shifters) to optimize **noise figure** and **insertion loss**.

TECHNICAL PROJECTS

Wi-Fi Propagation & Path Loss Simulator

- Built a physics-based **ray-tracing** simulation algorithm using **NumPy** to model wireless signal decay via the Log-Distance Path Loss model, accounting for frequency bands (2.4/5GHz) and environmental clutter.
- Utilized **Matplotlib**, enabling users to design custom floor plans and visualize signal coverage through heatmaps.

Your Body Guardian – Autonomous Pill Dispensing System

- Designed and prototyped an award-winning **microcontroller-based** pill dispensing system in a four-person team, earning **1st** place out of **~100 students** for technical execution and communication.

Analog Temperature Sensor Conditioning PCB

- Designed and simulated (**Altium/LTspice**) an analog signal-conditioning 2-layer PCB using a TMP36 sensor, incorporating a 5V→3.3V LDO, op-amp gain stage, RC low-pass filtering, and unity-gain buffering.